

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech, Winter 2025-26 Examination

Semester: VI

Subject Code: 303105363

Subject Name: Cloud Computing

Date: 19-11-2025

Time: 10.30 am to 1.00 pm

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. i. Define a hypervisor and its role in virtualization. ii. Explain the functions performed by a hypervisor in a virtualized environment. iii. Evaluate the impact of not having a hypervisor in virtualization.	06	CO3	BT2
	B. i. List the different ways cloud computing services are offered. ii. Provide real-world examples of each type of cloud service. iii. Evaluate the key differences between these types of cloud services.	06	CO1	BT2
Q.2	A. List and briefly explain the steps involved in setting up an IaaS environment.	04	CO4	BT1
	B. Compare traditional on-premise development environments with PaaS-based development.	05	CO1	BT4
	OR			
	B. What are the key characteristics of Web 2.0?	05	CO4	BT1
Q.3	A. What is DNS resolution time, and why is it important in cloud networking?	04	CO4	BT4
	B. How do cloud providers use machine learning and artificial intelligence to enhance security and protect data?	05	CO5	BT2
	OR			
	B. How does IBM Bluemix handle load balancing and auto-scaling?	05	CO4	BT2

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. i. Explain the core principles of securing a private cloud architecture. ii. Identify the critical security measures that need to be implemented in a private cloud environment. iii. Evaluate the effectiveness of these security measures in protecting the private cloud from threats.	06	CO5	BT1
	B. i. Identify the strategies businesses use to ensure high availability of data in the cloud.	06	CO5	BT4

	ii. Explain how redundancy, failover mechanisms, and geographic distribution contribute to high availability. iii. Evaluate the impact of high availability on business continuity and customer satisfaction in cloud environments.			
Q.2	A. Discuss the role of cloud service providers in managing cloud platforms.	04	CO4	BT2
	B. What is IaaS? Explain the significance of resource virtualization in IaaS.	05	CO4	BT1
	OR			
	B. Differentiate between Type 1 and Type 2 hypervisors.	05	CO3	BT6
Q.3	A. How does Web 2.0 impact data privacy in SaaS?	04	CO4	BT2
	B. Discuss the steps to configure a virtual machine (VM) using VMware Cloud.	05	CO5	BT2
	OR			
	B. What is cloud orchestration? Give an example.	05	CO1	BT2